

DRAKE FIELD, AR, USA

WMO: 723445

Lat: 36.010N

Lon: 94.169W

Elev: 381

StdP: 96.83

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93993

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.0	-8.9	-17.2	0.9	-9.8	-14.1	1.2	-6.1	11.1	13.0	9.9	11.5	2.4	0	0.457

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	35.1	23.5	33.4	23.8	32.3	23.7	25.7	31.2	25.1	30.8	24.5	30.0	3.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.1	19.9	28.5	23.5	19.1	28.0	22.8	18.4	27.4	81.3	31.4	78.6	30.7	76.0	30.2	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.4	8.4	7.6	-16.1	36.7	3.4	2.2	-18.5	38.3	-20.5	39.6	-22.4	40.8	-24.9	42.4
			-16.6	26.6	3.3	0.8	-18.9	27.2	-20.8	27.6	-22.7	28.0	-25.1	28.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.4	2.6	4.9	9.3	14.3	18.8	23.3	25.6	25.2	20.9	14.8	8.9	3.9	(6)
(7)		DBStd	9.32	6.18	6.10	5.66	4.68	4.06	2.79	2.59	2.85	3.88	4.74	5.27	5.45	(7)
(8)		HDD10.0	794	239	163	82	14	1	0	0	0	0	12	82	201	(8)
(9)		HDD18.3	2208	487	377	283	136	44	2	0	0	19	128	284	447	(9)
(10)		CDD10.0	2417	12	20	59	143	275	400	483	472	328	162	49	13	(10)
(11)		CDD18.3	786	0	0	3	14	59	152	225	214	97	19	2	0	(11)
(12)		CDH23.3	7828	0	4	24	120	489	1449	2372	2217	958	185	9	0	(12)
(13)		CDH26.7	3193	0	1	2	15	106	520	1099	1061	358	31	0	0	(13)
(14)	Wind	WSAvg	2.9	3.3	3.4	3.6	3.8	3.0	2.6	2.2	2.1	2.0	2.6	3.2	3.1	(14)
(15)	Precipitation	PrecAvg	1097	55	65	100	108	130	123	73	96	115	91	88	72	(15)
(16)		PrecMax	1753	147	171	221	203	266	212	180	232	264	244	243	174	(16)
(17)		PrecMin	534	12	30	25	34	32	44	7	12	19	1	24	19	(17)
(18)		PrecStd	256	38	35	53	47	62	49	55	58	69	76	62	44	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.5	22.6	26.0	28.3	31.1	34.1	37.2	37.6	34.8	29.1	24.6	20.6	(19)
(20)			MCWB	14.2	14.6	16.6	18.7	22.0	23.5	23.3	23.4	22.6	21.0	17.5	15.6	(20)
(21)		2%	DB	17.9	19.5	23.3	26.3	29.1	32.4	34.8	35.7	32.2	27.5	22.1	18.0	(21)
(22)			MCWB	13.5	13.6	15.6	17.6	21.3	24.0	23.9	23.4	22.3	19.8	15.8	14.5	(22)
(23)		5%	DB	15.9	17.5	21.3	24.5	27.7	31.1	33.1	33.7	30.7	25.5	20.0	16.1	(23)
(24)			MCWB	11.7	12.7	14.5	16.8	21.0	23.6	24.3	23.5	22.2	18.4	15.2	12.1	(24)
(25)		10%	DB	12.9	15.7	18.9	22.8	26.3	29.7	32.2	32.2	28.6	23.4	17.9	13.3	(25)
(26)			MCWB	9.0	11.4	13.3	16.2	20.2	23.0	24.3	23.4	21.4	17.8	13.9	9.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.7	17.3	18.4	20.7	24.1	25.9	26.6	26.4	24.8	22.8	19.5	17.3	(27)
(28)			MCDWB	18.9	20.0	23.4	25.7	28.0	31.0	31.8	31.6	30.8	27.5	23.2	19.1	(28)
(29)		2%	WB	14.9	15.6	17.1	19.6	23.0	25.1	25.8	25.4	23.9	21.6	17.7	15.2	(29)
(30)			MCDWB	17.3	18.4	21.6	24.0	27.3	30.5	31.3	31.3	29.8	25.6	20.0	17.9	(30)
(31)		5%	WB	12.4	13.8	15.7	18.7	21.9	24.3	25.3	24.8	23.2	20.3	16.2	13.1	(31)
(32)			MCDWB	14.5	16.8	19.9	22.8	26.2	29.2	30.9	30.8	28.3	23.5	19.1	15.4	(32)
(33)		10%	WB	9.7	11.6	14.4	17.7	21.1	23.7	24.8	24.1	22.3	18.9	14.6	10.7	(33)
(34)			MCDWB	13.0	14.7	18.3	21.5	25.3	28.4	30.2	30.1	26.9	22.3	17.4	12.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.5	11.8	12.5	12.8	11.4	11.0	11.4	12.2	12.8	13.0	12.2	11.1	(35)
(36)			MCDBR	14.5	14.8	14.9	14.5	12.4	12.3	12.9	14.3	14.4	14.7	14.2	14.1	(36)
(37)			MCWBR	9.9	10.0	8.7	7.9	6.2	5.3	4.6	4.6	5.8	7.7	8.9	10.1	(37)
(38)		5% WB	MCDBR	11.7	12.4	12.2	11.2	10.5	10.9	11.2	11.6	11.9	11.2	11.5	11.7	(38)
(39)			MCWBR	9.7	9.7	8.4	7.3	5.9	5.3	4.9	4.6	5.4	6.7	8.8	9.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.304	0.307	0.341	0.378	0.417	0.425	0.447	0.446	0.388	0.345	0.315	0.293	(40)	
(41)		taud	2.493	2.492	2.427	2.330	2.272	2.273	2.265	2.242	2.377	2.484	2.521	2.541	(41)	
(42)		Ebn at Noon	897	938	928	905	870	859	837	831	867	881	878	890	(42)	
(43)		Edh at Noon	82	92	108	125	135	135	135	135	112	92	79	74	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.47	2.99	4.04	5.08	5.62	6.40	6.35	5.79	4.94	3.69	2.66	2.12	(44)	
(45)		RadStd	0.22	0.41	0.24	0.29	0.49	0.59	0.39	0.39	0.37	0.48	0.30	0.27	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page